

HTML Multimedia & Advanced Elements

A beginner's guide to embedding video, audio, interactive UI elements, semantic tags, and image maps in HTML.

1. 🎬 Start the Lecture with Video

Instead of putting a link to a video on our webpage, HTML allows us to put the video directly inside the webpage.

```
<video controls width="600" poster="thumbnail.jpg">
  <source src="lesson.mp4" type="video/mp4">
  <source src="lesson.webm" type="video/webm">
  Your browser does not support video.
</video>
```

controls

Shows playback controls

width

Sets the video width

poster

Thumbnail image before play

<source>

Multiple formats for browser compatibility

fallback content

Shown if browser doesn't support video

🔥 **Challenge:** What happens if you remove controls? Try <video src="lesson.mp4"></video> — the browser's default playback controls disappear.

2. 🎧 Add an Audio Lecture



What if our lesson doesn't need video? Maybe we want students to listen to an explanation.

```
<audio controls>
  <source src="lecture.mp3" type="audio/mpeg">
  <source src="lecture.ogg" type="audio/ogg">
  Your browser does not support audio.
</audio>
```

<video>

Visual + Audio

<audio>

Audio only; supports controls and multiple <source> elements similarly to <video>

🎓 **Student Activity:** Create a webpage containing a lecture video, a background audio clip, a fallback message, and multiple media sources.

3. 💬 Introduce Video Captions

A video isn't only about video and sound. We can also attach text tracks.

```
<video controls width="600">
  <source src="lesson.mp4" type="video/mp4">
  <track src="captions.vtt" kind="subtitles"
         srclang="en" label="English">
</video>
```

```
WEBVTT

00:00:01.000 --> 00:00:04.000
Welcome to our HTML lecture.

00:00:04.000 --> 00:00:07.000
Today we are learning multimedia.
```



- **<track>** works with WebVTT for subtitles/captions
- Makes video content more accessible

4. 📖 <details> — Hidden Information

How can we hide information until the student wants to see it?

```
<details>
  <summary>What is HTML?</summary>
  <p>HTML stands for HyperText Markup Language.</p>
</details>
```

The browser automatically gives you an expandable section. Make it a quiz:

```
<details>
 <summary>
```

5. 🪟 <dialog> — Create a Popup

```
<dialog open>
 <h2>
```



6. 📊 <progress> — Learning Progress



```
<h3>Your HTML Progress</h3>
<progress value="70" max="100">70%</progress>

<p>Course completion:
  <progress value="7" max="10"></progress>
</p>
```

- Turns the lecture into a learning dashboard
- 🔥 **Ask students:** What happens if you change 7 to 9?

7. 📈 <meter> — Skill Level

```
<p>HTML Skill:
  <meter min="0" max="100" value="80">80%</meter>
</p>

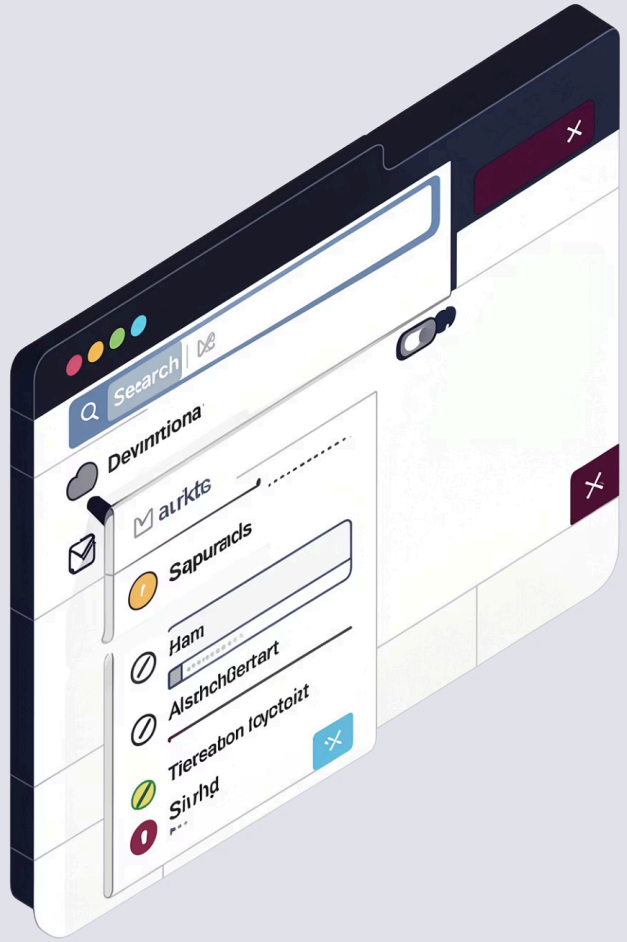
<p>Battery:
  <meter min="0" max="100" value="25">25%</meter>
</p>
```

<progress>	<meter>
Completion of a task	Measurement within a known range

8. 🔎 <datalist> — Smart Input

```
<label for="browser">Choose your favorite browser:</label>
<input list="browsers" id="browser">
<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
  <option value="Safari">
</datalist>
```

- Students can type or select a suggestion
- **<datalist>** provides predefined suggestions associated with an input — a tag many beginners don't know!



9. 🧮 <output> — Display a Result

HTML can also provide a place where the result of a calculation can appear.

```
<form>
  <label>Marks:
    <input type="number" value="80">
  </label>
  <output>80%</output>
</form>
```



- JavaScript can dynamically update <output>
- Natural introduction: **HTML** → Structure | **JavaScript** → Behaviour

10. ⌨️ <kbd> — Keyboard Instructions



```
<p>Press <kbd>Ctrl</kbd> + <kbd>S</kbd> to save your file.</p>

<p>Press <kbd>Ctrl</kbd> + <kbd>C</kbd> to copy.</p>
```

- **<kbd>** represents keyboard input — particularly useful when teaching coding

11. 💻 <code>, <pre> and <samp>

```
<pre><code>
<!DOCTYPE html>
<html>
 <body>
  <h1>Hello World</h1>
 </body>
</html>
</code></pre>

<p>Browser output: <samp>Hello World</samp></p>
```



<code>

Computer code



<pre>

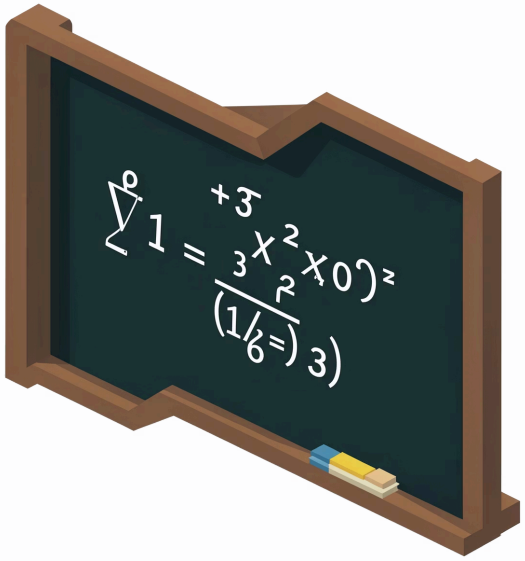
Preserves formatting



<samp>

Sample program output

12. 🔬 <var> — Variables



```
<p>Area = <var>length</var> × <var>width</var></p>

<p>x = 10</p>
```

- **<var>** represents a variable in mathematics or programming

13. 🌍 <abbr> — Abbreviations

```
<p>
  <abbr title="HyperText Markup Language">HTML</abbr>
  is used to structure webpages.
</p>
```



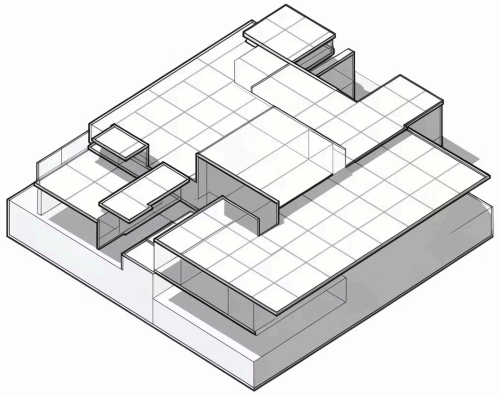
- Ask students to hover over **HTML** — the full title appears as a tooltip

14. 🖍️ <mark> — Highlight Important Information

```
<p>
  The most important topic today is
  <mark>HTML Semantics</mark>.
</p>
```

> <mark>
>
> **<mark>** is useful when you want to highlight text that is relevant to the current context

15. 🖼️ <figure> + <figcaption>



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#1B1B27 #3C3933

```
<figure>
  <img src="html-structure.png"
       alt="HTML document structure" width="400">
  <figcaption>
    Figure 1: Basic HTML document structure
  </figcaption>
</figure>
```

- Gives students a proper way to associate an image with its caption

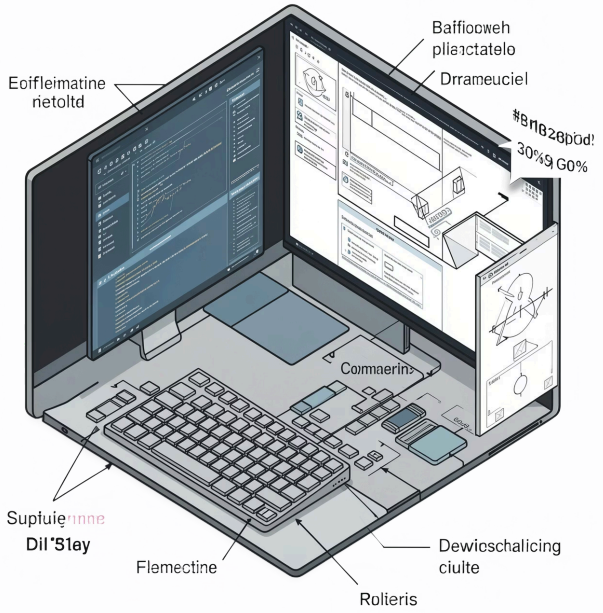
16. 🖼️ <picture> — Responsive Images

```
<picture>
  <source media="(max-width: 600px)" srcset="mobile.jpg">
  <source media="(min-width: 601px)" srcset="desktop.jpg">
  <img src="desktop.jpg" alt="Learning illustration">
</picture>
```

- 🔥 **Ask students:** Which image will appear on a mobile screen?
- Students can experiment by resizing the browser



17. 🗺️ Image Map — Make an Image Interactive




```
<img src="computer.png" alt="Computer"
    usemap="#computerMap" width="500">

<map name="computerMap">
  <area shape="rect" coords="50,50,200,200"
      href="https://example.com" alt="Monitor">
</map>
```

- A particular region of an image becomes clickable
- 🎓 **Fun assignment:** Create an interactive image map of your own!